AI-Based Internet Performance Analysis and Forecasting of Starlink Connectivity in Oman

Department of Computer Science

German University of Technology in Oman

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14 June 2026

Presented by: Leen Mohammed Amer Al Kayyal (22-0024)

Overview

1

Measure

2 residential sites

2

Compare

Starlink vs local fibre

3

Analyze

Latency, jitter, loss, download, upload

4

Forecast

Next 15-minute latency

5

Deploy

Dashboard + advisor

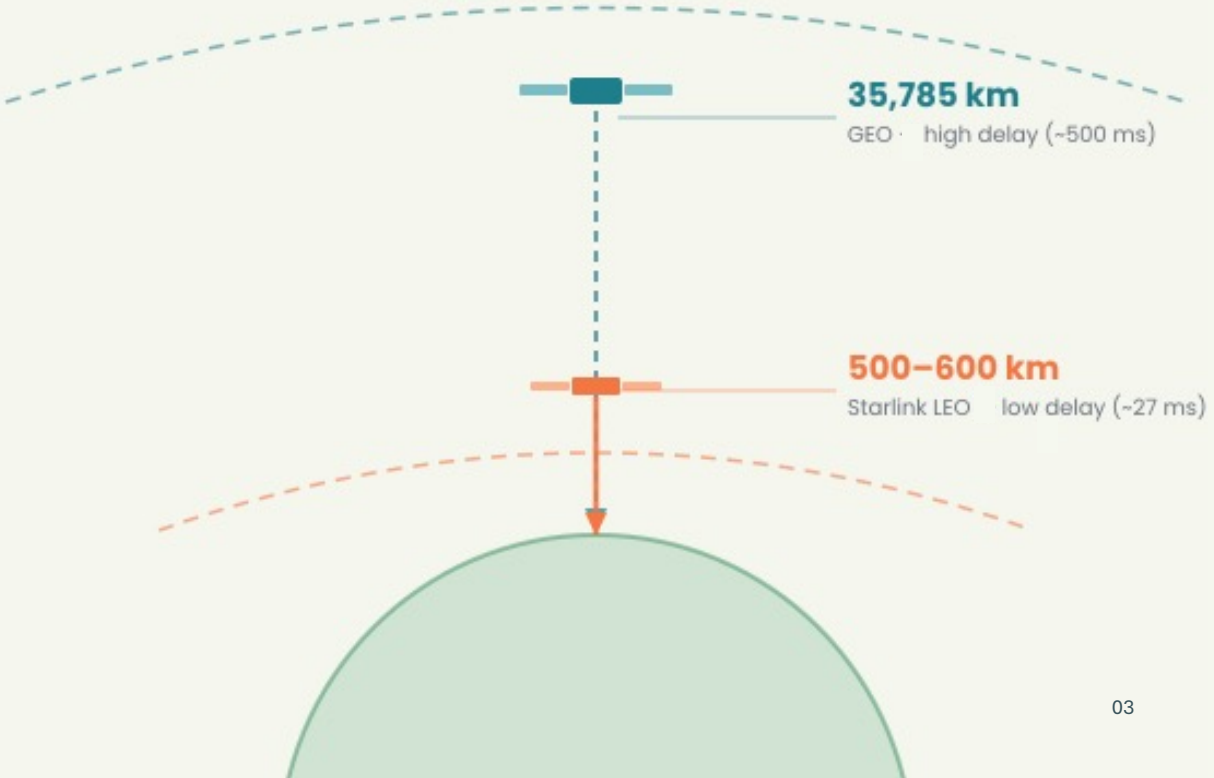
Why Starlink Matters in Oman?

What is Starlink?

- SpaceX satellite internet, no cables, no towers

The problem with old satellites and Why Starlink is different

- GEO at 35,785 km → 500 - 700 ms delay
- LEO at 500–600 km → ~27 ms delay



Sources: SpaceX (2024); Cakaj (2021); Britannica (2025); ITU-T G.114.

Literature & Research Gap

01 **No published Oman study**

Almost all studies come from Europe or North America

02 **No time-matched comparison**

No side-by-side measurement against Omani providers

03 **No local forecasting model**

No ML model trained on Omani Starlink measurements

04 **No decision-support tool**

No advisor translating metrics into practical user guidance

Sources: Thesis literature review; Michel et al. (2022); Mohan et al. (2024); Tiwari et al. (2023).

Research Questions

01	**Performance**	How does Starlink perform in real Muscat homes?
02	**Comparison**	How does it compare with local providers?
03	**Decision support**	Can short-term latency forecasting support user decisions?

Experimental Design

EXPERIMENT A

STARLINK vs Omantel

Al Mawaleh South, Muscat

07 Mar - 28 Mar 2026

Target: 2,057 observations

Interval: 15 minutes, 24/7

Starlink Gen 3 terminal

EXPERIMENT B

STARLINK vs AWASR

Al Hail South, Muscat

20 Apr - 12 May 2026

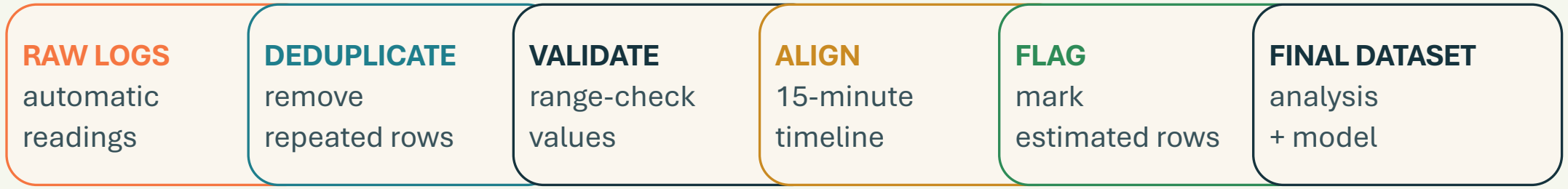
Target: 2,097 observations

Interval: 15 minutes, 24/7

Starlink Gen 3 terminal

Source: Thesis Ch. 3, experimental design and deployment windows.

Data pipeline



Dataset	Expected Slots	Raw Rows	Rows Removed	Real Obs.	Estimated Rows
Starlink Data A	2,057	2,052	0	1,979	64
Omantel Data A	2,057	**14,472**	12,427	1,429	628
Starlink Data B	2,097	2,085	1	2,067	30
Awasr Data B	2,097	**21,620**	19,525	1,281	816

Source: Thesis Ch. 3, data cleaning and preprocessing methodology.

Backup: 36

Dataset quality

After cleaning, the four datasets held 6,756 real measured observations out of almost 40,000

Real Data Availability



4,154

final 15-minute slots across both experiments

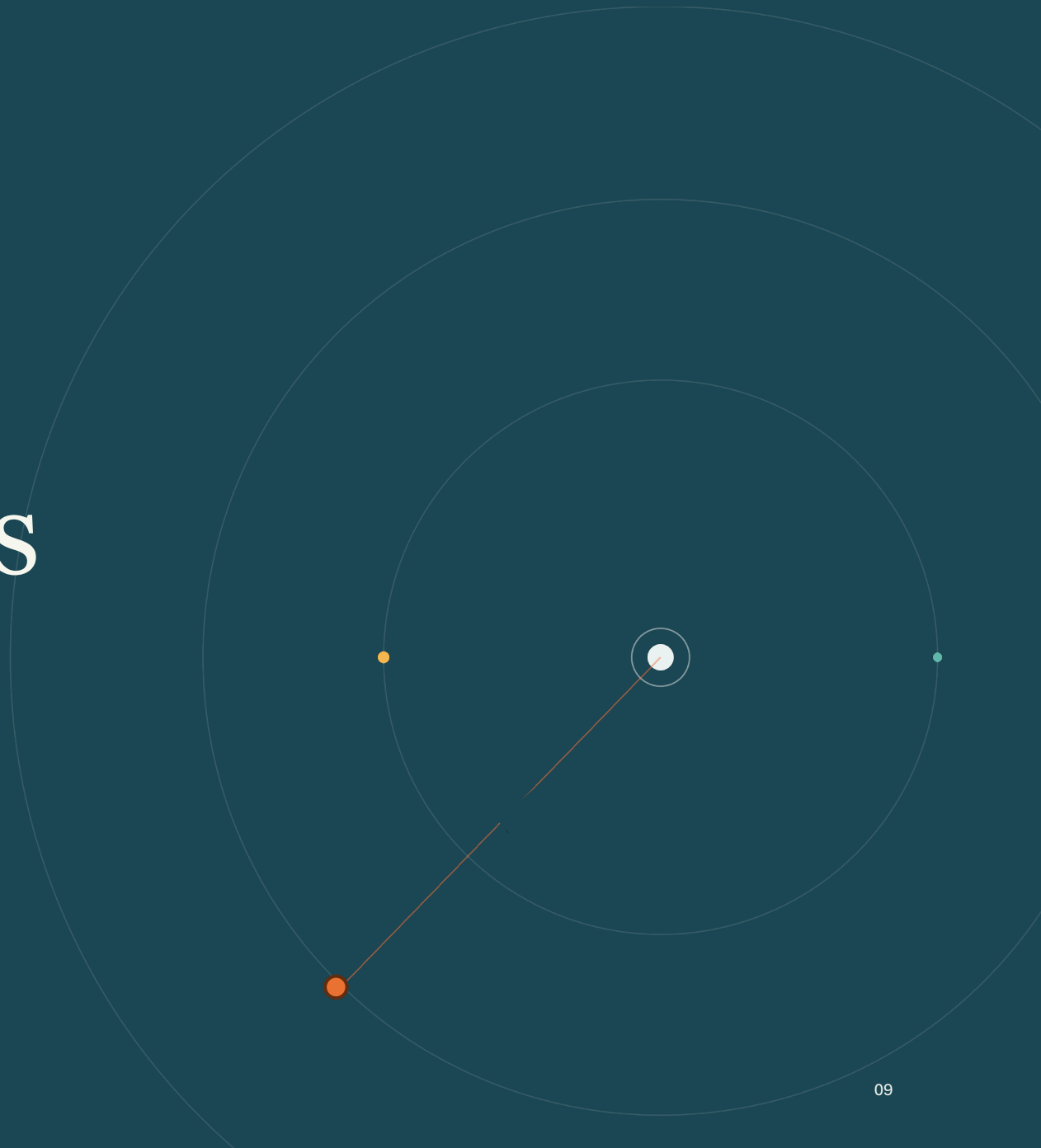
4,050

usable forecasting rows after feature engineering

Source: Thesis Ch. 4 and Appendix A, preprocessing summary and forecasting dataset.

Backup: 35–36

Results & Analysis



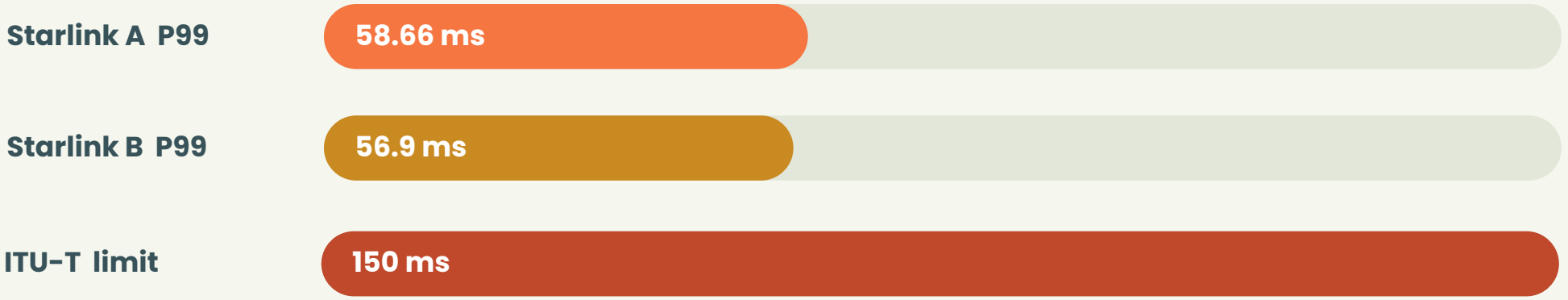
Latency

27.48 ms
Starlink A
Median latency

25.80 ms
Starlink B
Median latency

5.90 ms
Omantel A
Median latency

4.43 ms
Awasr B
Median latency

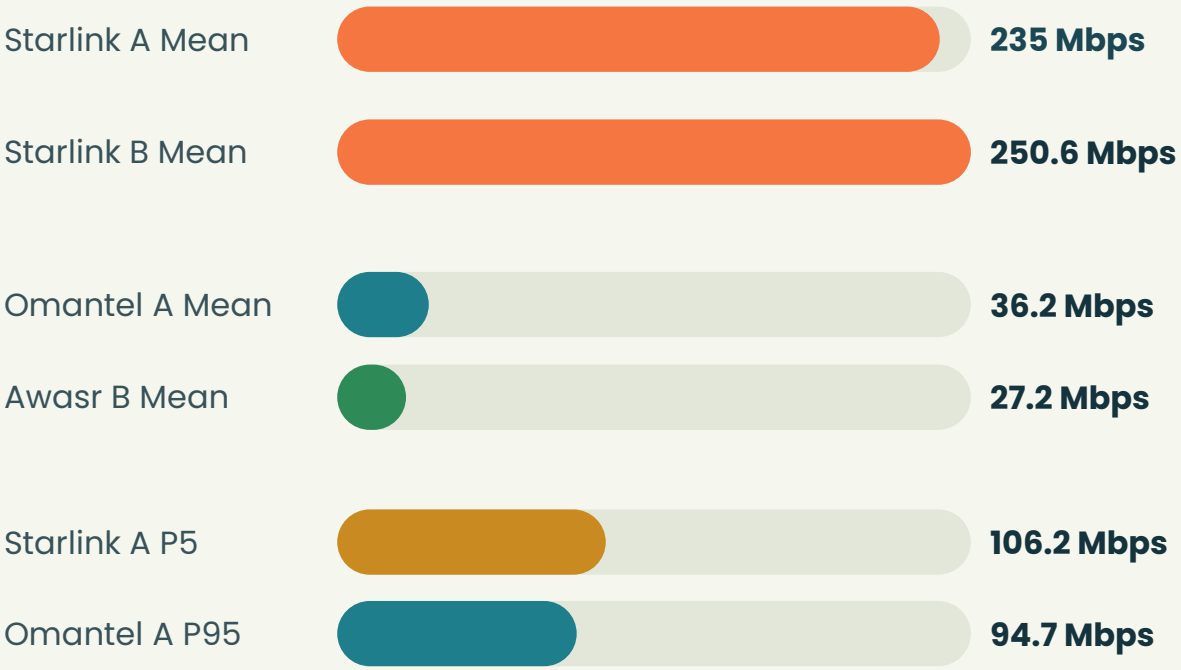


ITU-T = International Telecommunication Union
G.114 = delay guideline for real-time/interactive communication
150 ms = accepted benchmark

Source: Thesis Ch. 4, Tables 3 and 5; ITU-T G.114 latency guidance.

Throughput

DOWNLOAD SPEED (Mbps)



UPLOAD SPEED (Mbps)

Starlink A		Omantel A	
Mean	34.9	Mean	55.0
P5	15.5	P5	2.6
Starlink B		**Awasr B**	
Mean	35.3	Mean	4.0
P5	15.9	P5	1.0

Source: Thesis Ch. 4, Tables 3 and 5.

Backup: 43

Application suitability

≥ 90%
70–89%
<70%

Application	Starlink A	Omantel A	Starlink B	Awasr B
Web Browsing	Reliable	Reliable	Reliable	Reliable
Video Call (Zoom/Teams)	Reliable	Reliable	Reliable	Marginal
HD Streaming	Reliable	Reliable	Reliable	Reliable
Online Gaming	Reliable	Reliable	Reliable	Reliable
Large File Upload / Backup	Reliable	Marginal	Reliable	Unreliable
4K Video Conferencing	Reliable	Unreliable	Reliable	Unreliable

Source: Thesis Ch. 4, Tables 4 and 6; ITU-T G.114 thresholds.

Backup: 37

What Explains Variability?

01 **Time of day**

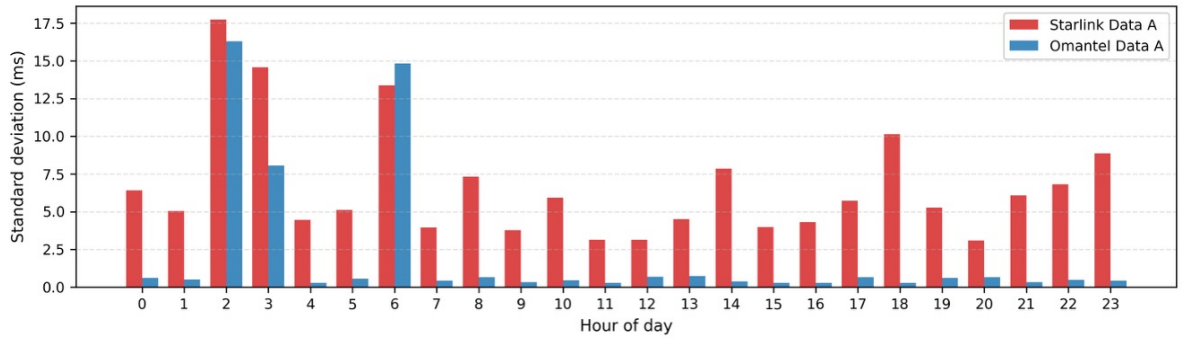
Significant effect, but spikes spread across all hours, not only evenings

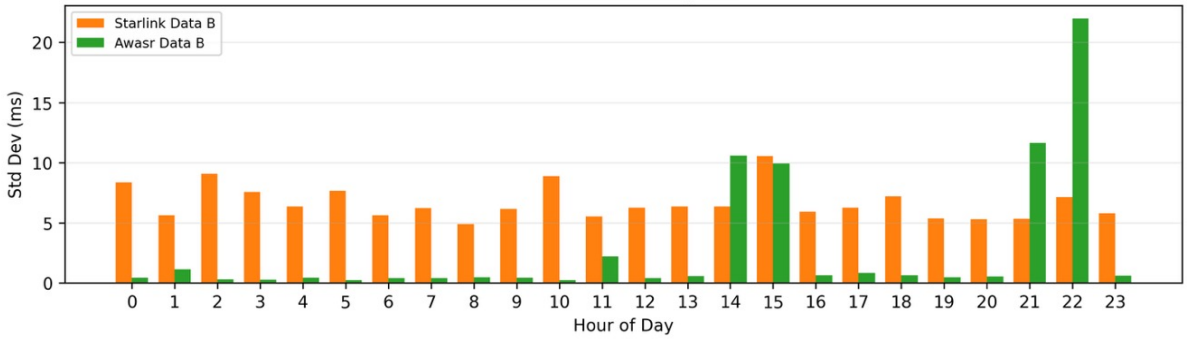
02 **Weather**

Weak effects, and none significant for latency

03 **Internal Starlink network behavior**

Routing, gateway and handover decisions are invisible from terminal-side data







Source: Thesis Ch. 4, Figures 4, 6, and 10; Mohan et al. (2024); Izhikevich et al. (2024).

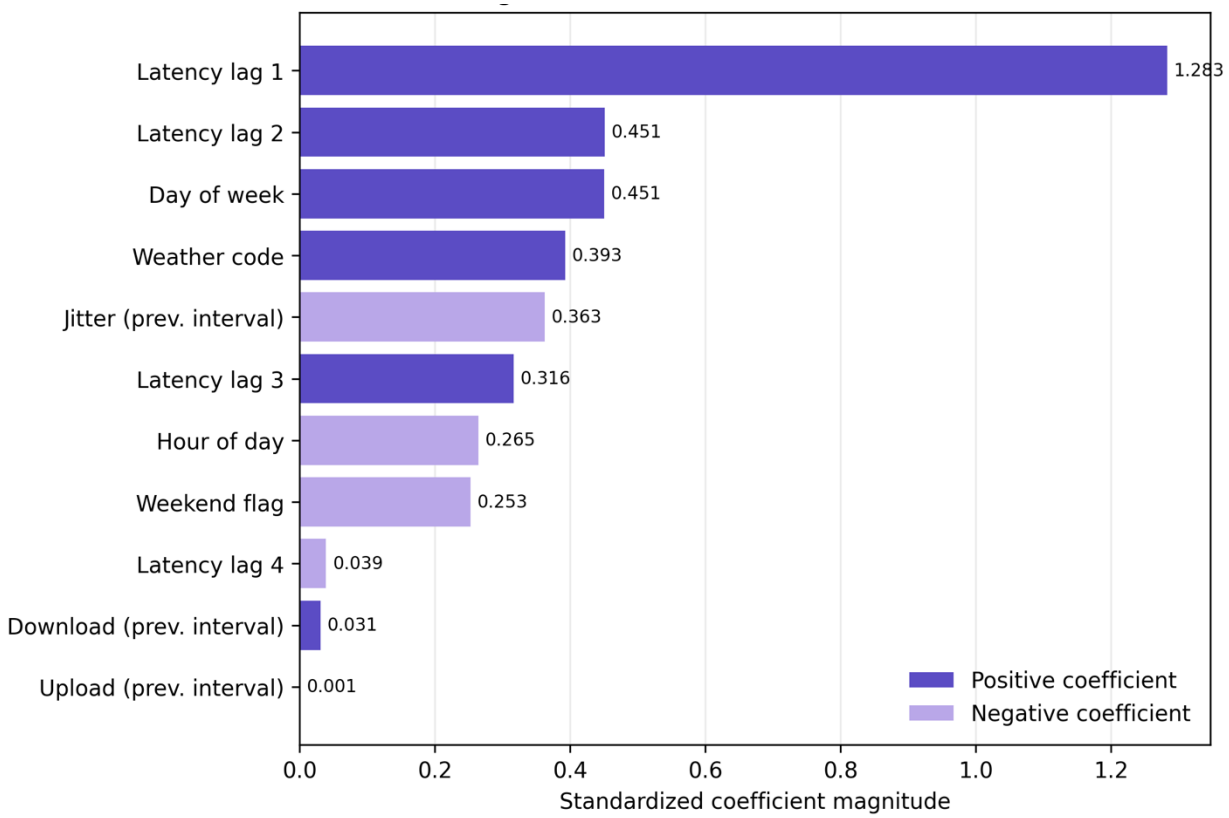
Backup: 38, 46

Forecasting: Setup & Features

SETUP

→ **Predict the next 15-minute latency** → **Chronological train / test split**

FEATURES



Source: Thesis Ch. 3, Appendix B, and Thesis Ch. 4 Figure 20; Ferreira et al. (2023); Saha et al. (2024).

Backup: 39

Forecasting Results & Cross-Site Check

Naive baseline

5.95

MAE

Linear Regression

4.34

MAE
RMSE 6.47

Random Forest

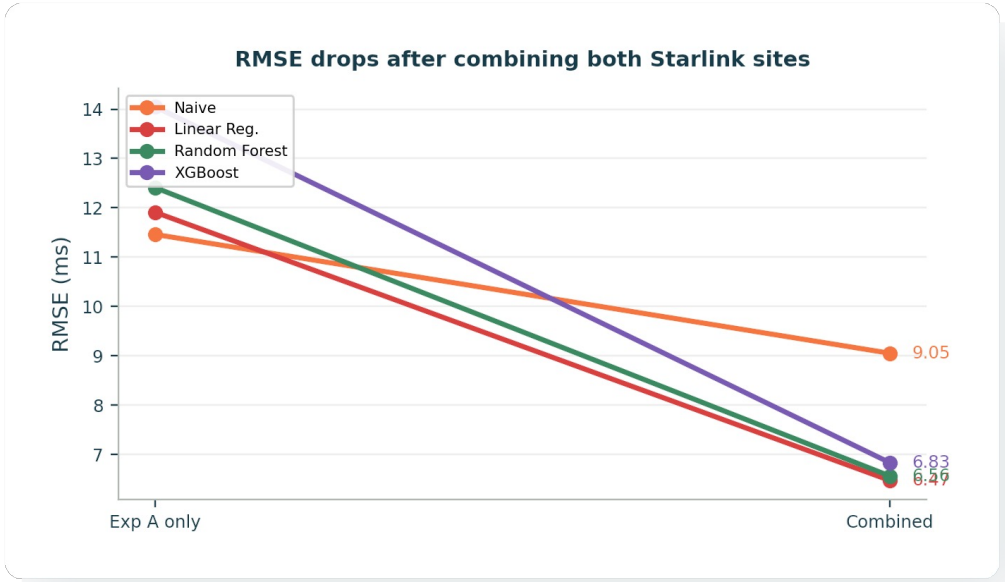
4.52

MAE

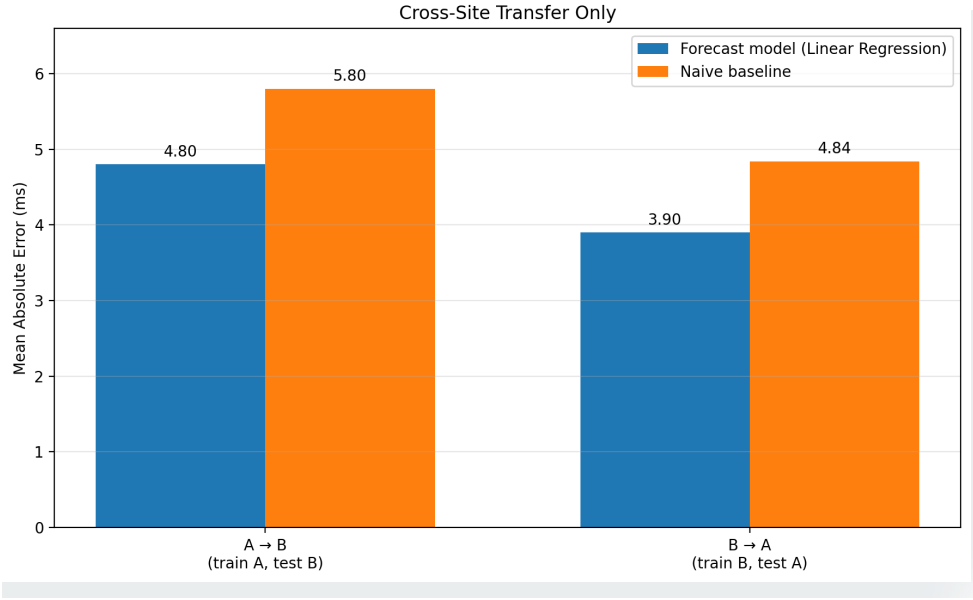
XGBoost

4.63

MAE



45.7% **lower RMSE vs Experiment A alone**



Extra validation: one-site training

Source: Thesis Ch. 4, forecasting model results; additional cross-site transfer check.

Backup: 40

Robustness & Explainability

Setup	Model	MAE (ms)	RMSE (ms)	R^2
Lag 1-4	Linear Regression	**4.337**	**6.466**	-0.008
Lag 1-12	Linear Regression	**4.302**	**6.621**	-0.080
LSTM sequence 12	LSTM	**4.369**	**6.457**	-0.023

The LSTM was trained on a sliding window of 12 latency observations

Initial check of forecast-based latency classification

- Correct for normal readings, but rare elevated-latency cases were not detected

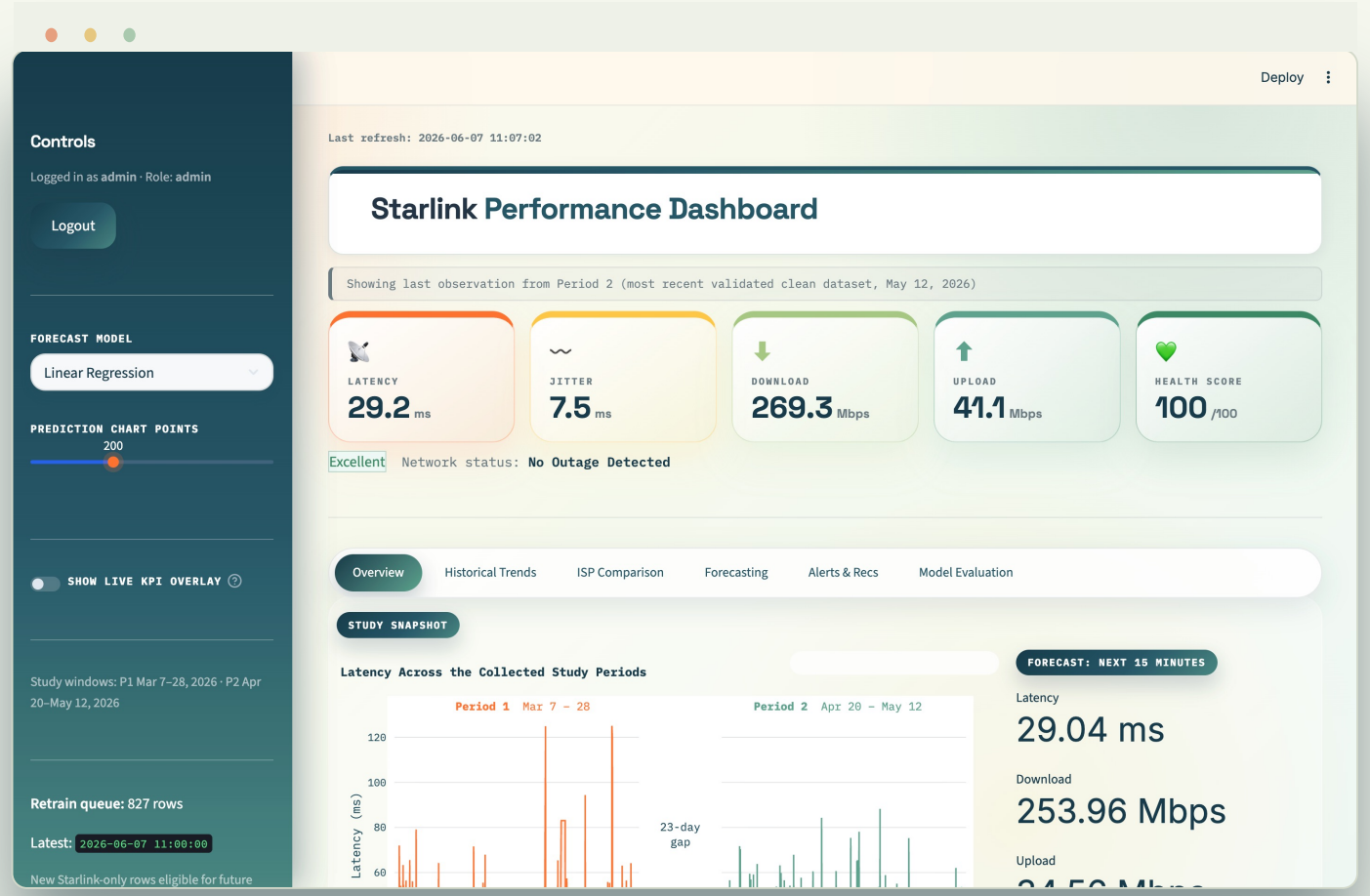
98.9%

overall accuracy on an imbalanced test set

Source: Thesis Ch. 4, Sections 4.7–4.9.

Backup: 40, 44

Monitoring Dashboard



Source: Thesis dashboard implementation, Appendix C.

00 **Overall View**
full dashboard layout

01 **Live Monitoring**
current status

02 **Historical Trends**
latency, jitter, throughput

03 **ISP Comparison**
Starlink vs terrestrial

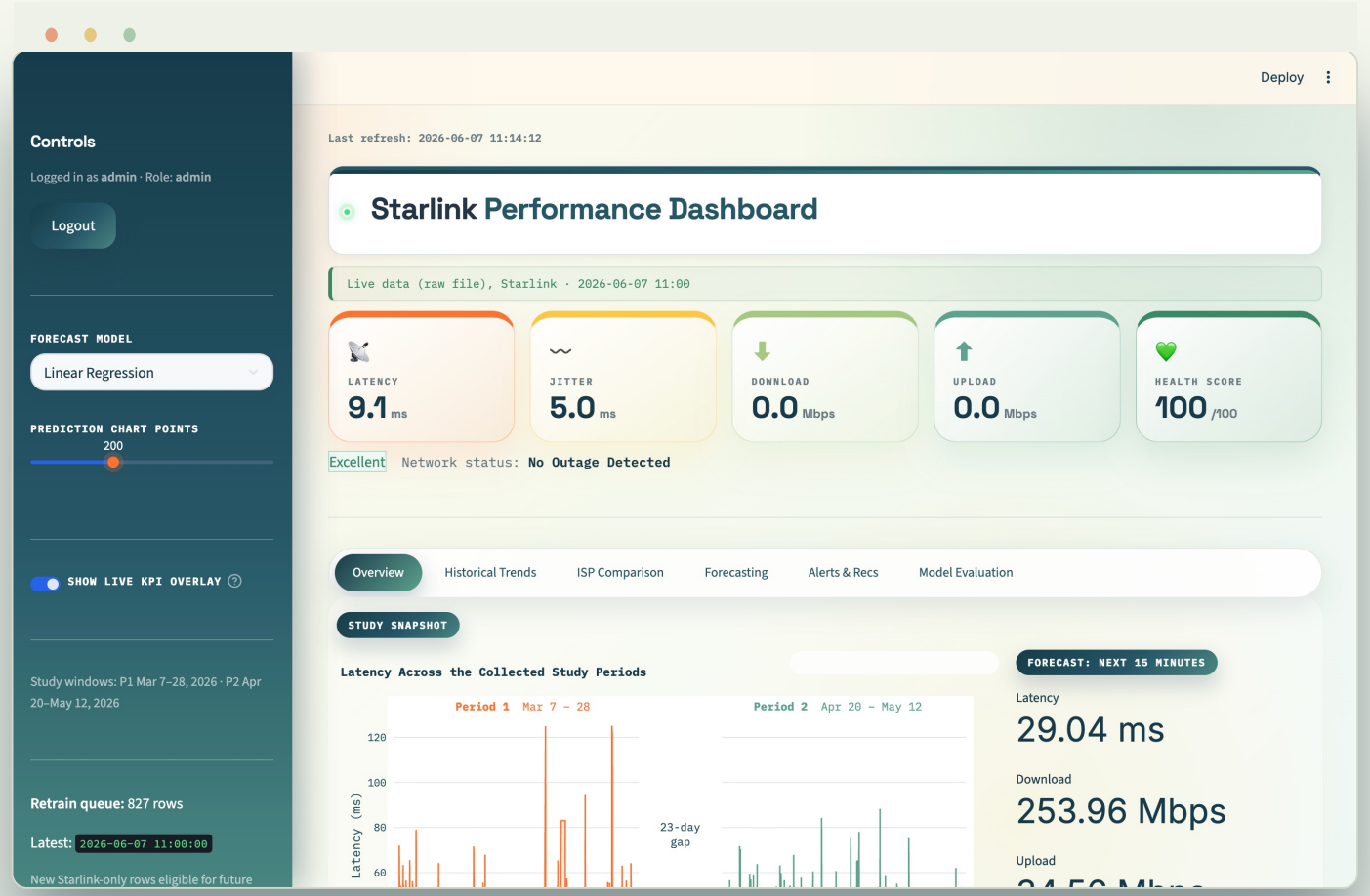
04 **Forecasting**
next-interval prediction

05 **Alerts & Recommendations**
suggested actions

06 **Model Evaluation**
accuracy & error metrics

Backup: 42

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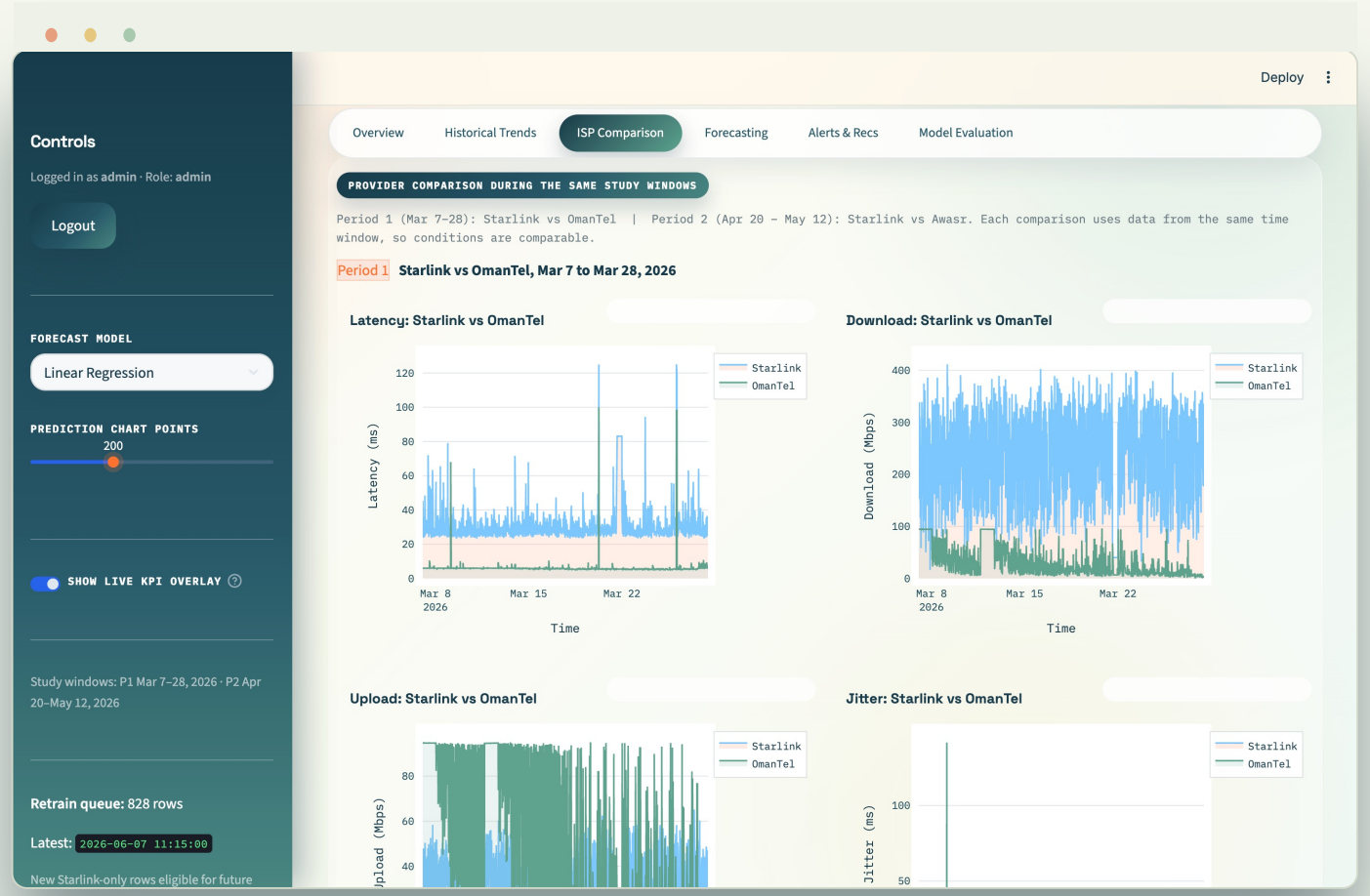
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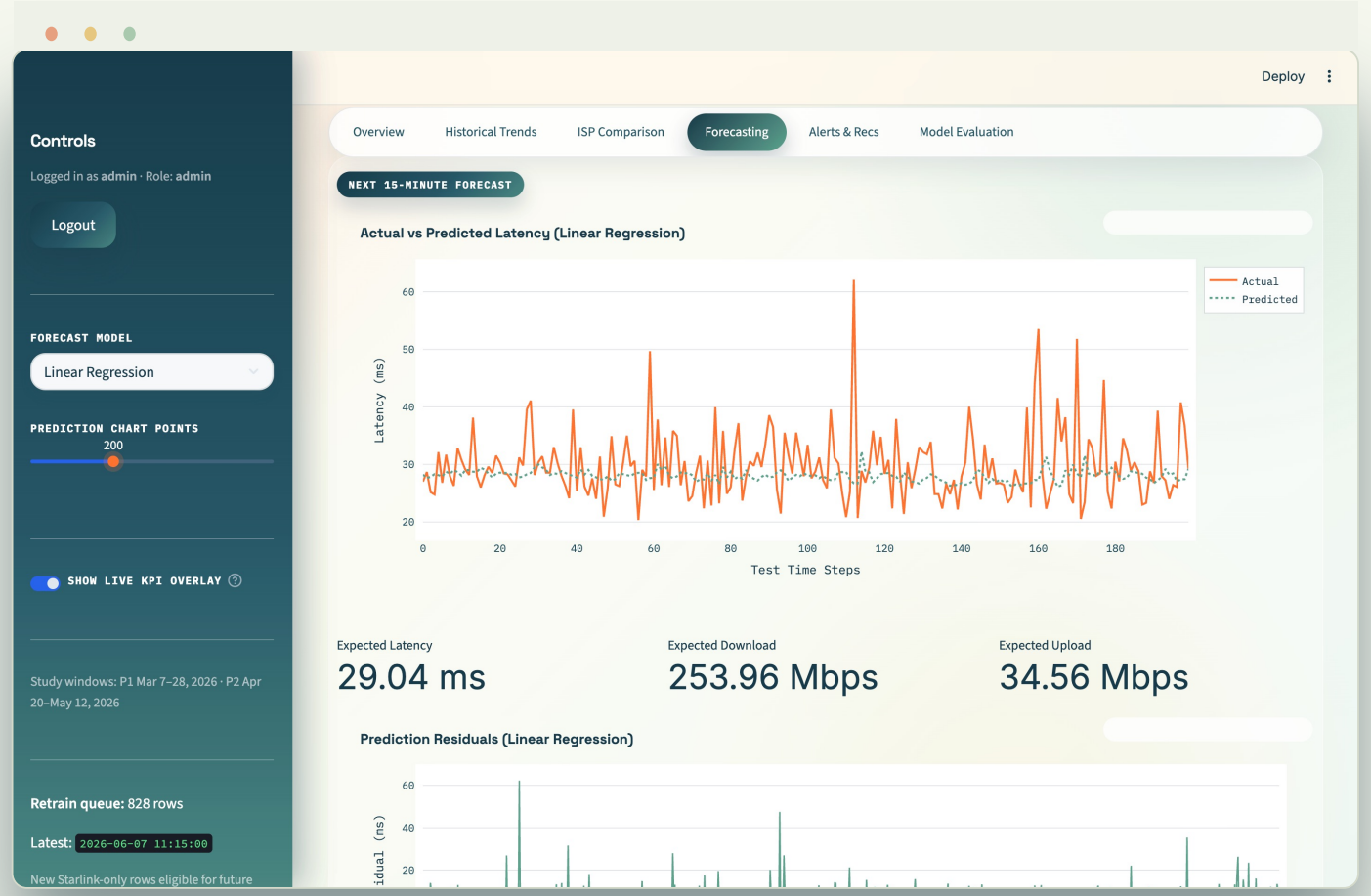
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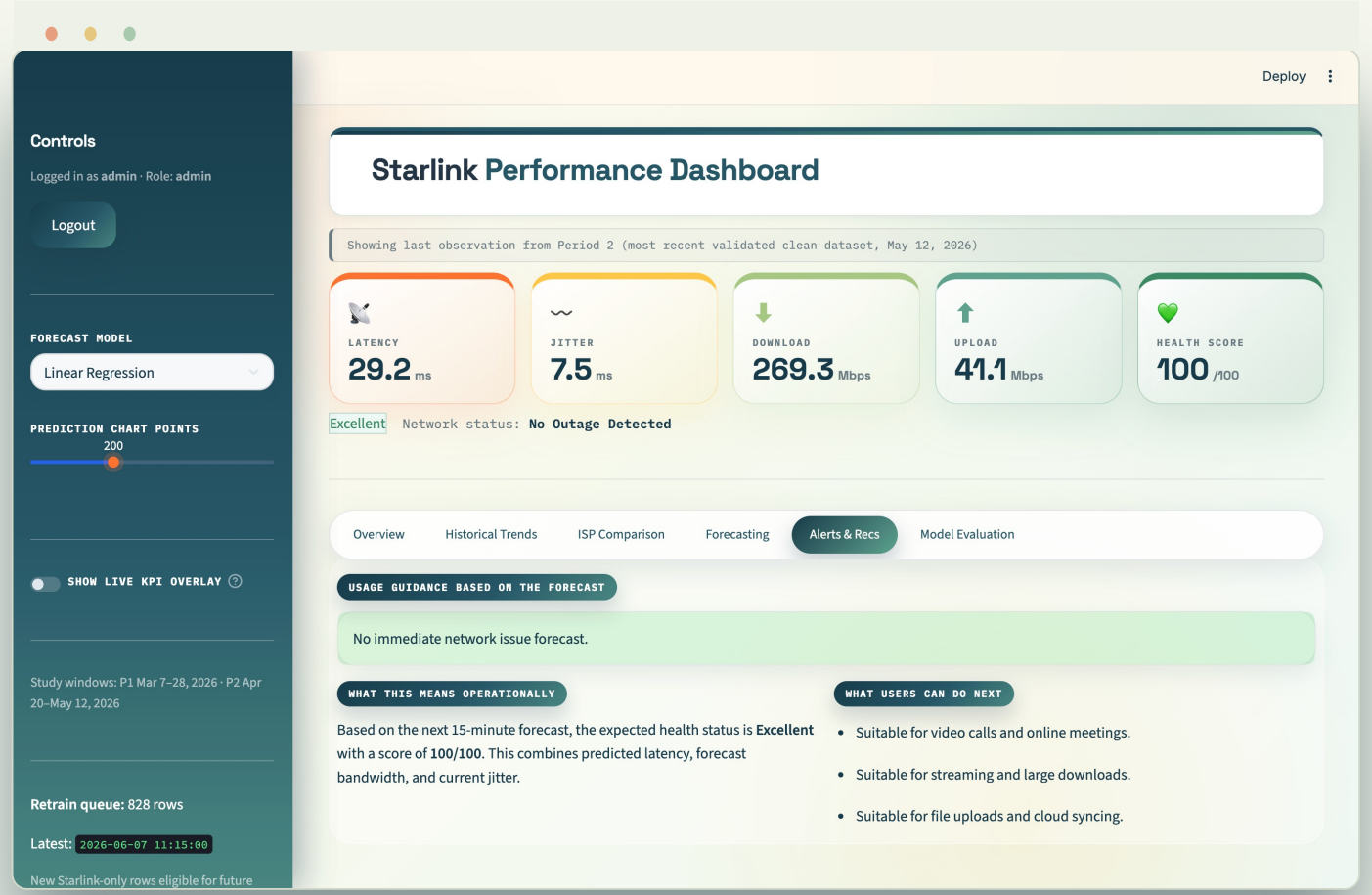
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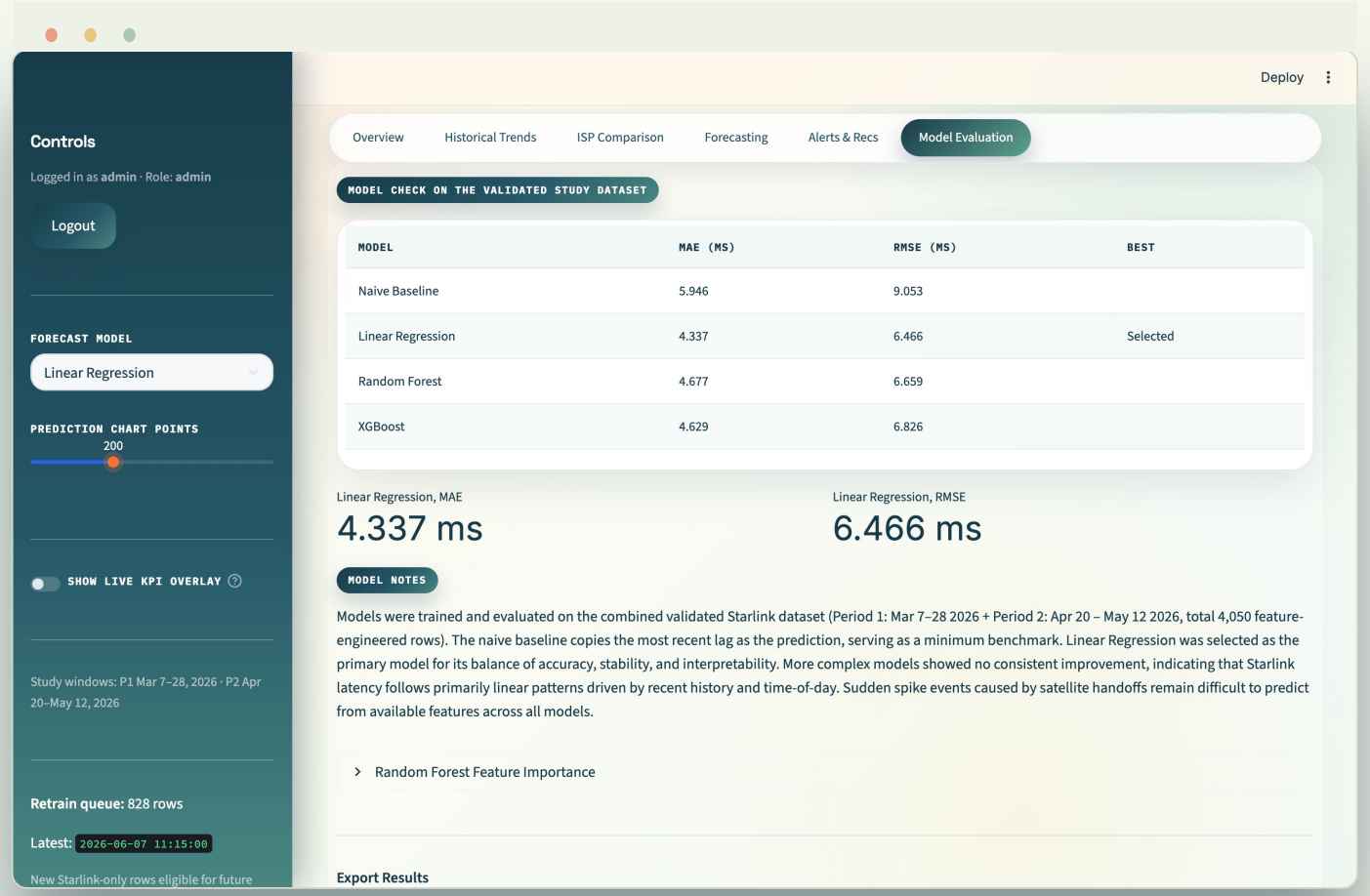
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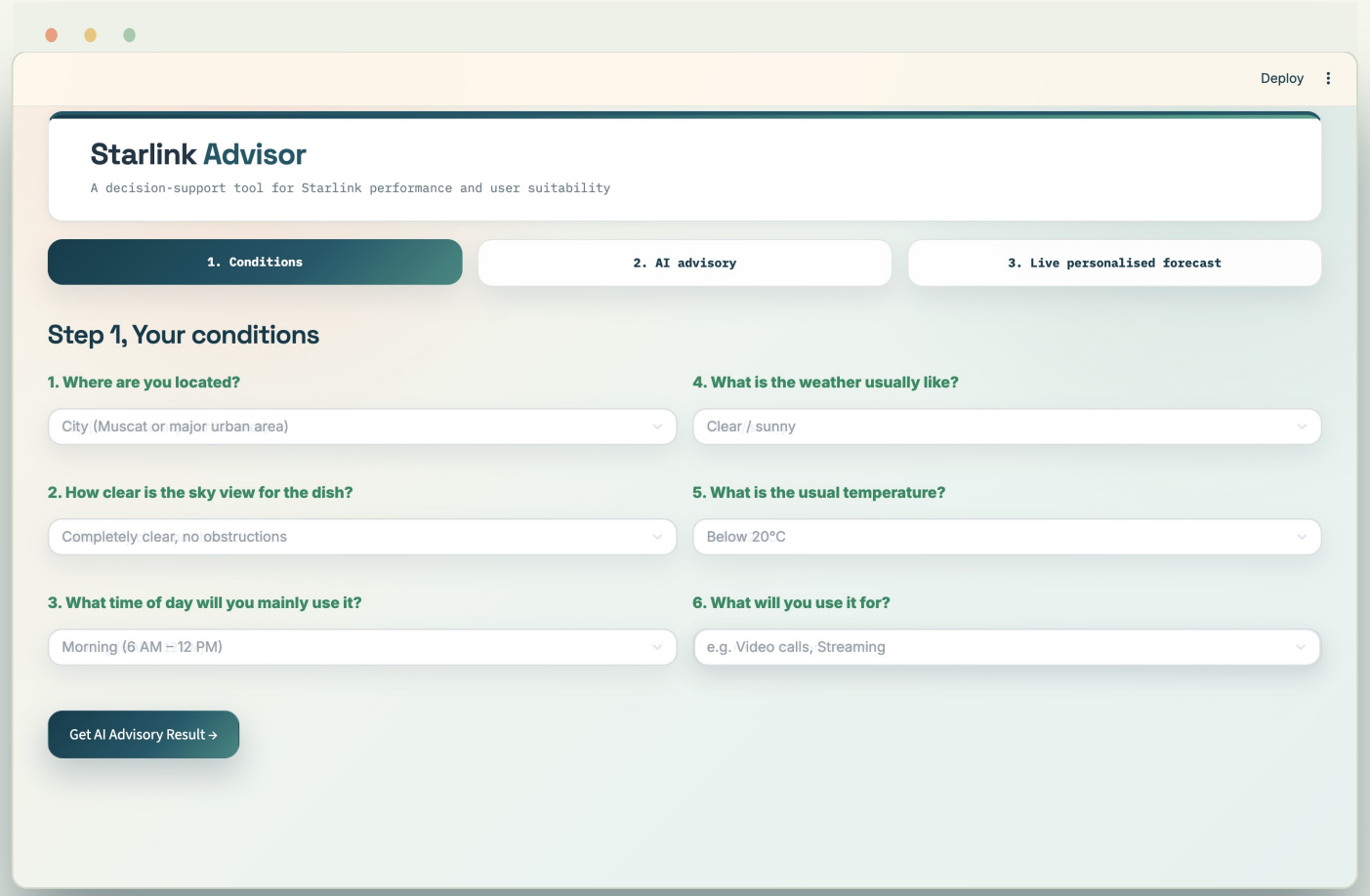
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Starlink Suitability Advisor



01 **Your Conditions**
location, sky view, usage

02 **AI Advisory Result**
forecast + use-case ratings

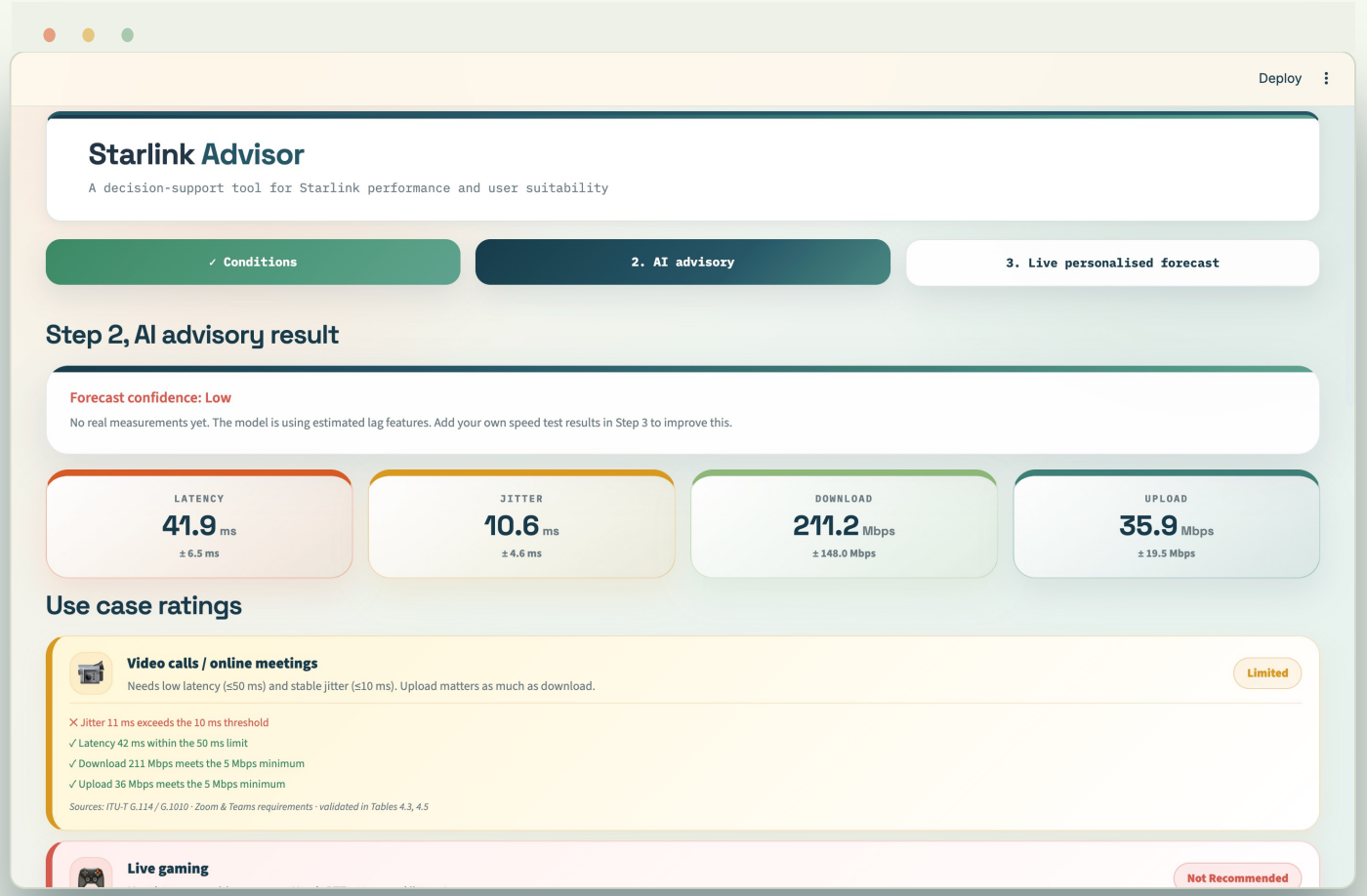
03 **Add Measurements**
upload CSV or enter manually

04 **Live Personalised Forecast**
refined from your real data

Source: Thesis advisor implementation, Sections 3.15 and 4.9.

Backup: 41

Starlink Suitability Advisor



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location, sky view, usage

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forecast + use-case ratings

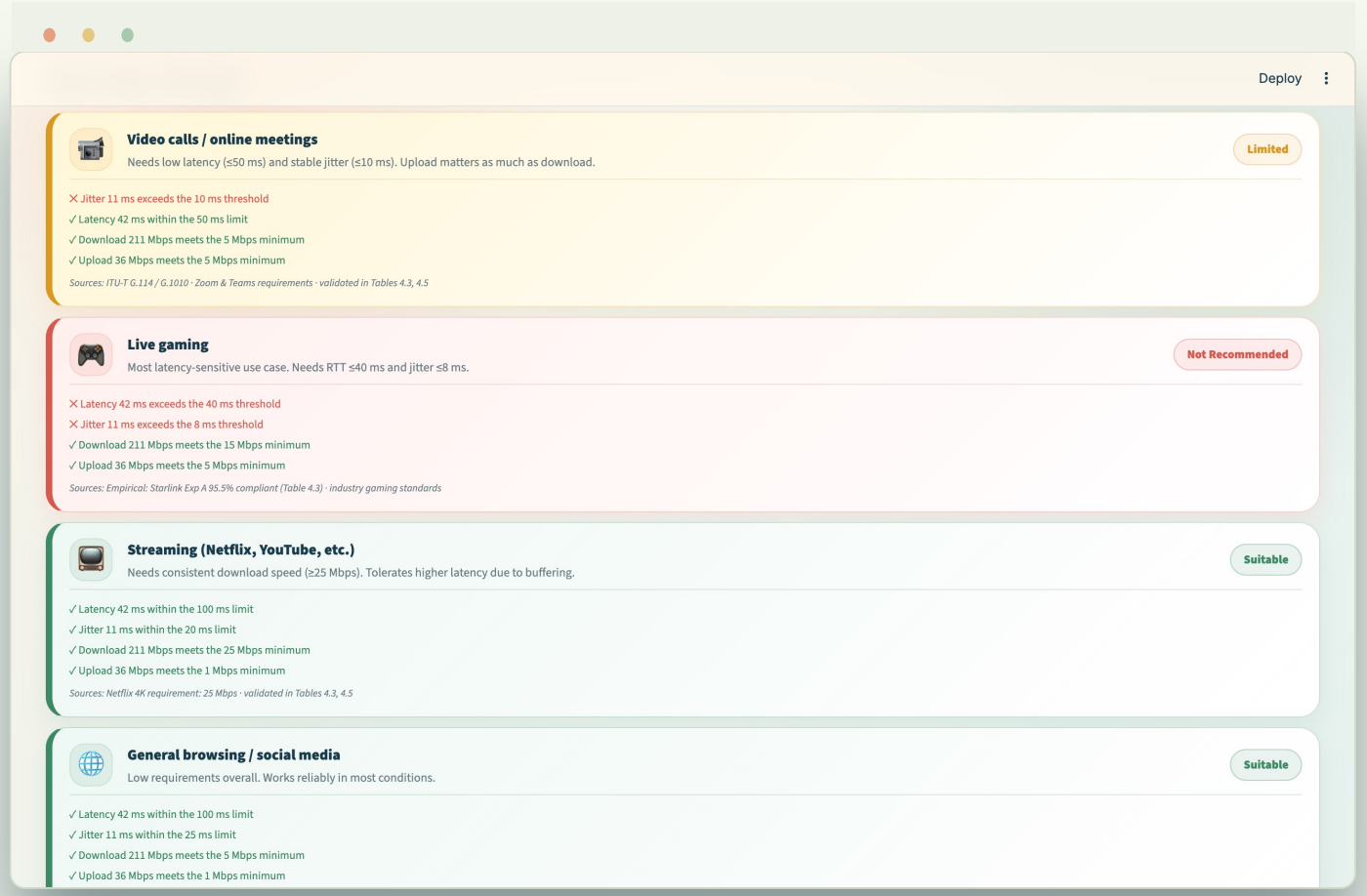
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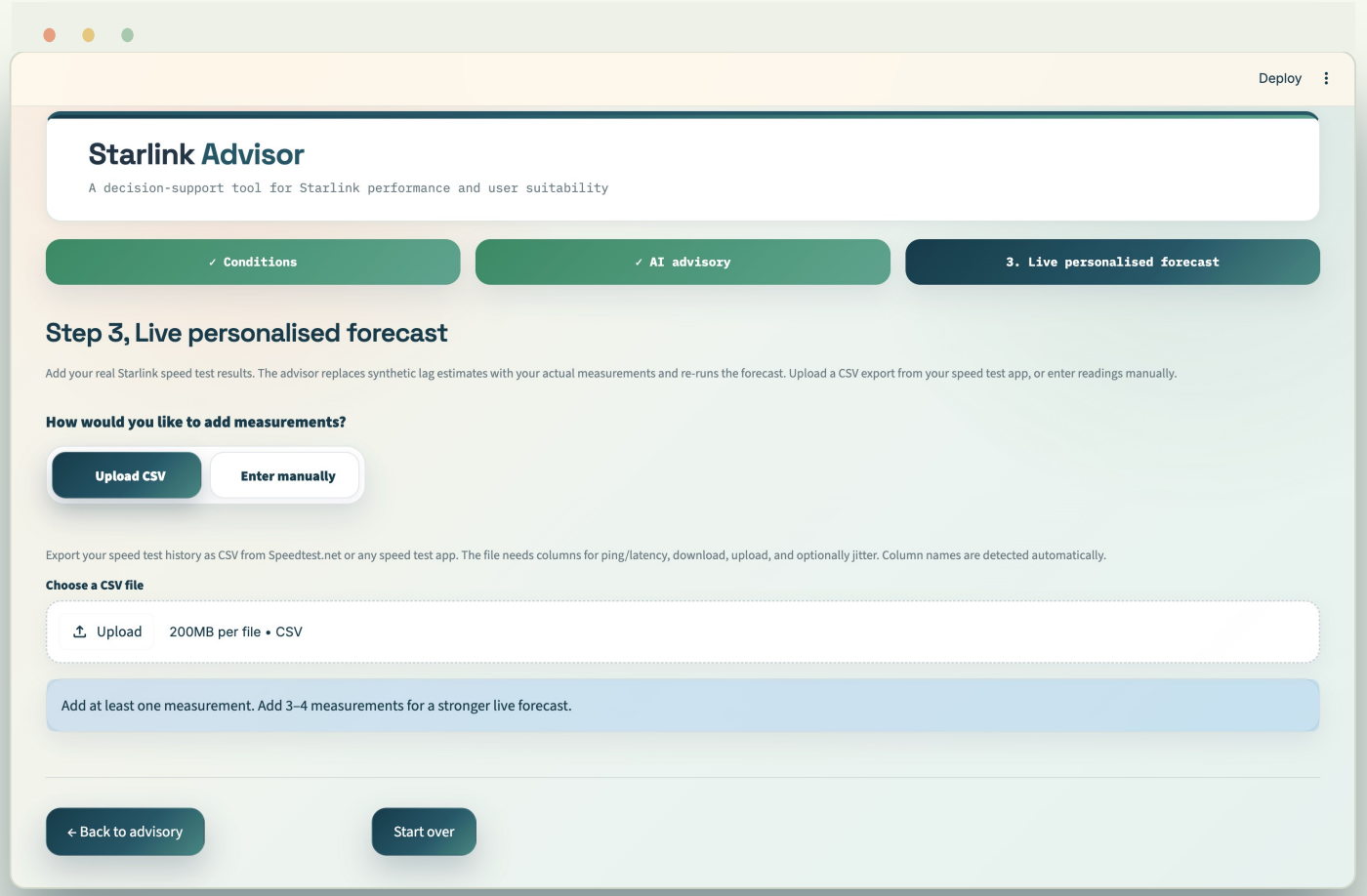
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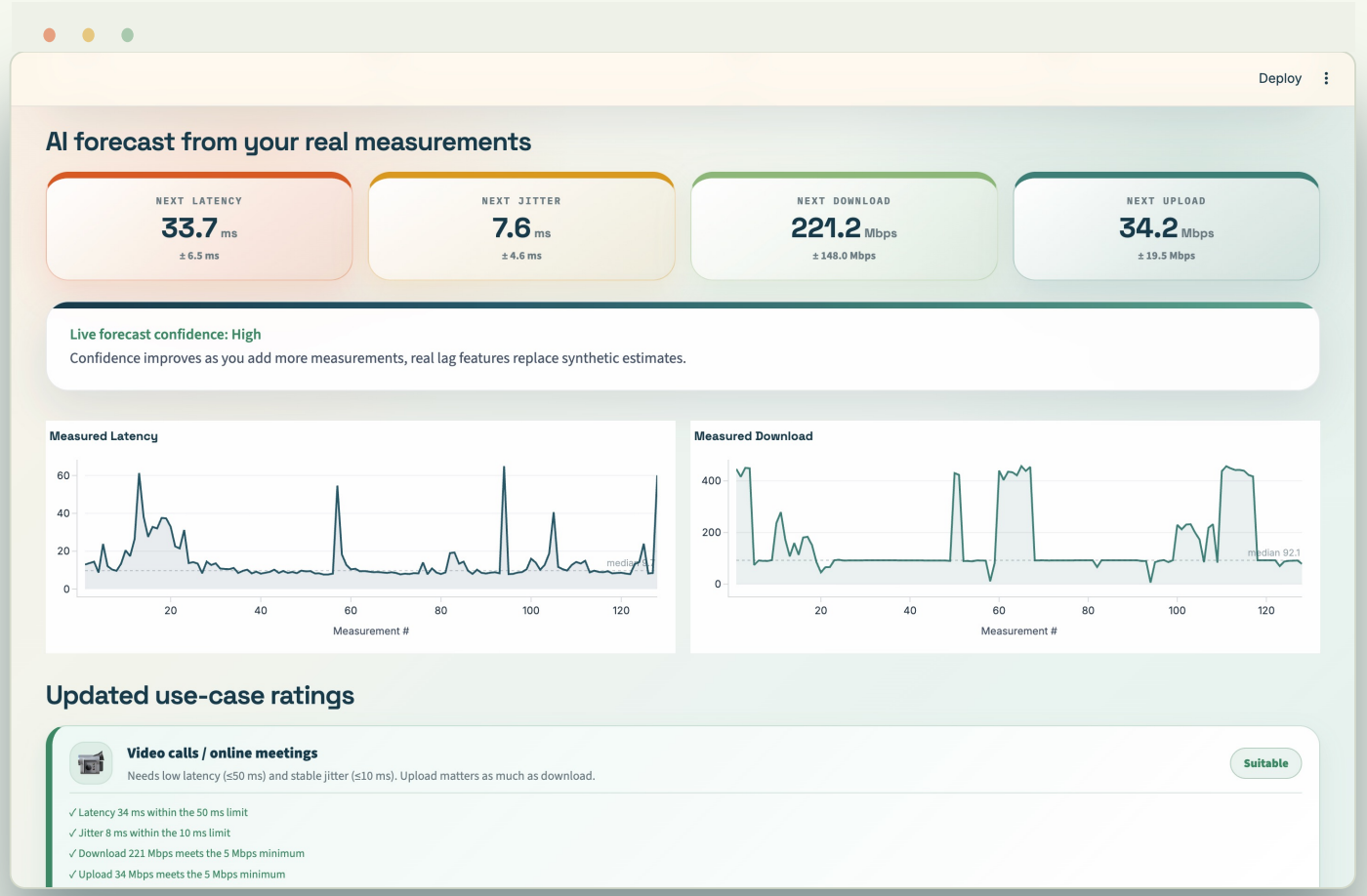
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My Contribution

The data and intelligence layer of the project.

01 Cleaned and prepared all four datasets

02 Ran the statistical and suitability analysis

03 Built the supervised learning datasets

04 Trained and compared the forecasting models

05 Integrated everything into the dashboard and advisor

Limitations & Future Work

LIMITATIONS

- Two close homes in Muscat
- Small dataset: 4,050 forecasting rows; few spike examples
- Data gaps: outage, rate limiting, duplicates
- Fibre download values affected by automated Speedtest setup
- Terminal-side data cannot see routing, gateway, or handover decisions

FUTURE WORK

- More regions: Nizwa, Salalah, coastal areas
- Longer runs across summer, winter, and Khareef
- More user-uploaded measurements to improve the advisor
- Gateway/routing-aware data if available
- Longer-range forecasting beyond the next 15 minutes
- Validate automated throughput using multiple servers and manual benchmark tests

Source: Thesis Ch. 5, limitations and future work.

Conclusion

25–28 ms

median latency, both sites

< 59 ms

P99 latency, both sites

6 / 6

application tests passed

4.34 ms

forecast error (MAE)



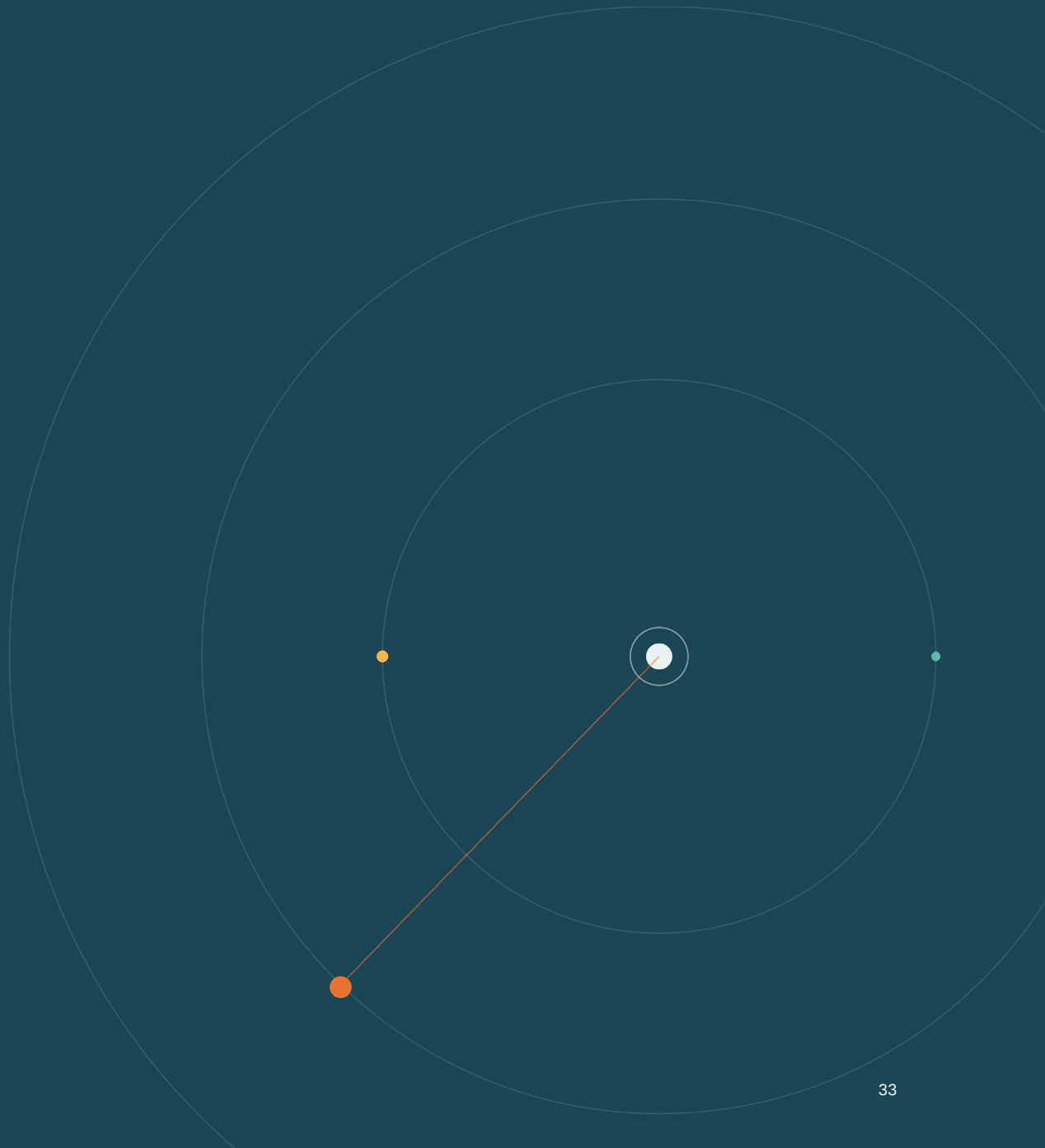
From raw measurements to practical connectivity guidance

In the measured homes, Starlink showed stable latency and credible backup potential.

Source: Thesis Ch. 5, conclusion summary.

Thank you

Questions & discussion



Special thanks to Dr. Ali for trusting me with this project and supporting me throughout it, Dr. Nesrine for helping shape the project, and Dr. Nabil and Dr. Nafaa for their guidance and support over the years.

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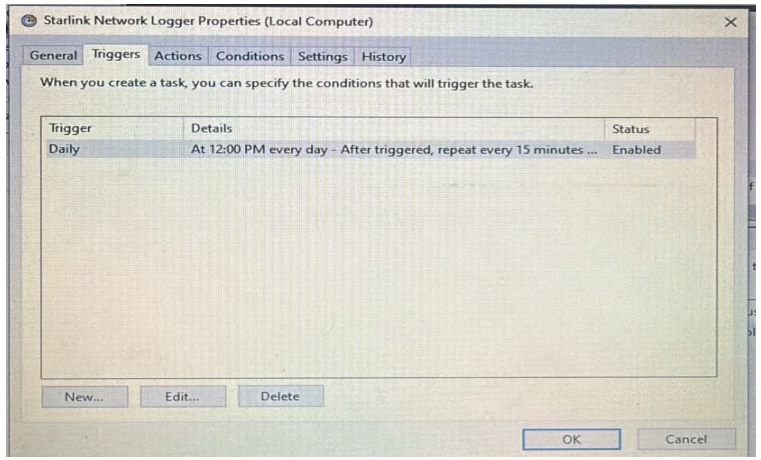
Tiwari, S., Bhushan, S., Taneja, A., Kassem, M., Luo, C., Zhou, C., He, Z., Raman, A., Sastry, N., Qiu, L., & Bhattacherjee, D. (2023). T3P: Demystifying low-Earth orbit satellite broadband. arXiv.

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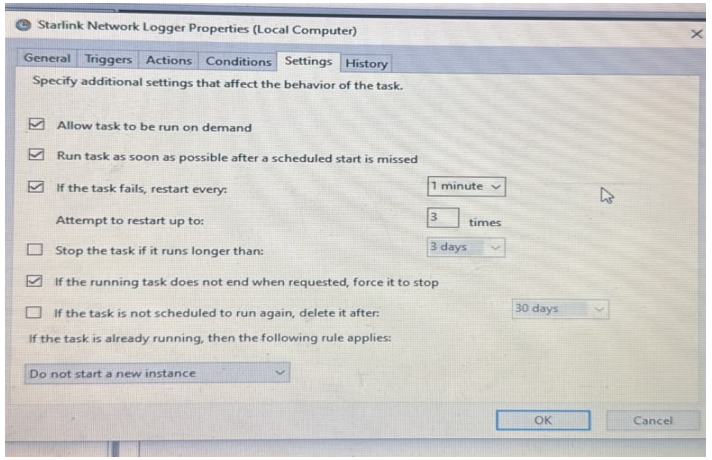
Appendix

Logger Evidence + Extra Rows

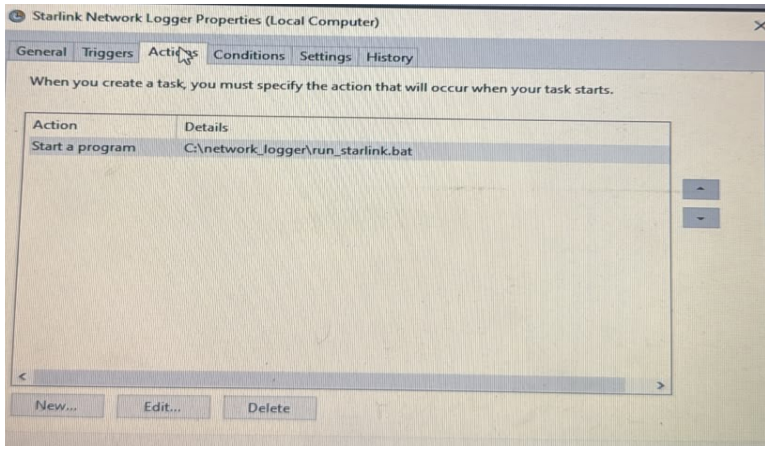
Task scheduler settings



Trigger: repeat every 15 min



Settings: restart/catch-up enabled



Action: run logger batch

Orow per script execution

```
# net_logger_starlink.py
st_ping, down, up, server_name, server_loc = run_speedtest()
...
append_row(LOG_FILE, row)
print("Logged:", ts)
```

- The script itself appends only one row when it runs.
- Therefore, extra rows mean extra executions, not extra valid 15-minute observations.
- Task Scheduler allowed missed-run catch-up and restart on failure every 1 minute up to 3 times.
- Fibre tests likely triggered this more often because the Speedtest server/path was less stable in the automated setup.
- This is reported as a likely logging/scheduler artefact, not a confirmed provider fault.

Cleaning

one provider → one record → one 15-minute slot

Formula used for slot logic

```
slot = floor(timestamp / 15 minutes) × 15 minutes
```

Handling rules

Raw case	Action
Exact duplicate	Remove repeated copy
Multiple rows in one slot	Keep one representative median row
Invalid / impossible value	Set missing or exclude from real analysis
Missing slot	Insert timestamp and flag
Estimated row	Timeline only, not ML training

Evidence in dataset/report

- Raw row count was not used as final sample size.
- Expected slots: 2,057 and 2,097.
- Estimated rows were flagged clearly.
- Analysis and ML used real measured rows only.

```
if was_estimated_row:
    keep for timeline/charts
    exclude from training/statistics

if multiple rows in slot:
    aggregate numeric values by median
```

Six-Application Suitability Logic

$$\text{Pass}_i(\text{app})=1 \text{ if } \text{Lat}_i \le \text{Lmax} \wedge \text{Jit}_i \le \text{Jmax} \wedge \text{Loss}_i \le \text{Pmax} \wedge \text{Down}_i \ge \text{Dmin} \wedge \text{Up}_i \ge \text{Umin}$$

Suitability percentage = number of rows that passed ÷ total real rows × 100

≥90% Reliable 70–89% Marginal <70% Unreliable

Code from advisor.py

```
if lat > uc["lat_max"]: score += 2
if jitter > uc["jitter_max"]: score += 1
if dl < uc["dl_min"]: score += 2
if ul < uc["ul_min"]: score += 2
color = "green" if score == 0 else "amber" if score <= 2 else "red"
```

Thresholds used for the six use cases (ITU-T G.114)

Use case	Lat≤	Jit≤	Down≥	Up≥
Web browsing	100	25	5	1
Video call	50	10	5	5
HD/4K streaming	100	20	25	1
Online gaming	40	8	15	5
Large upload/backup	150	30	5	10
4K video conferencing	50	10	25	15

Statistical Checks Conducted

Hourly latency

```
Kruskal–Wallis test
H=54.29, p=0.0002 (A)
H=79.01, p<0.0001 (B)
```

Tests whether latency distributions differ by hour.

95% CI by hour

```
CI = mean ± 1.96 × (s/√n)
```

Shows uncertainty around hourly average latency.

Weather effects

```
Spearman correlation
ρ with p-value
```

Checks monotonic relation between weather and metrics.

Spike / CDF checks

```
P95/P99 + empirical CDF
```

Shows worst-case behaviour, not only averages.

I tested time patterns, uncertainty, weather, and tail behaviour, so the conclusion is not based only on mean values.

AI Forecasting Model: What Actually Happens

Target y_t = latency at next 15-minute interval

Example: 10:00, 10:15, 10:30, 10:45 → predict 11:00

Feature engineering code

```
df["target"] = df[target_col].shift(-1)
for col in ["ping_avg_rtt_ms", "ping_jitter_ms", "download_mbps", "upload_mbps"]:
    for lag in range(1, 5):
        df[f"{col}_lag_{lag}"] = df[col].shift(lag)

features = hour + day_of_week + is_weekend + weather + metric_lags
```

Model setup

- Problem type: regression, not classification.
- Prediction horizon: one step ahead / 15 minutes.
- Split: chronological train/test to avoid leakage.
- Baseline: naive persistence, $\hat{y}_t = y_{t-1}$.
- Models: Linear Regression, RF, XGBoost, LSTM check.

Model Evaluation + Why Linear Regression Won

Evaluation formulas

- $MAE = mean(|y - \hat{y}|)$
- $RMSE = \sqrt{mean((y - \hat{y})^2)}$
- $R^2 = 1 - SSE/SST$
- Naive baseline: $\hat{y}_t = y_{t-1}$

Combined Starlink model results

Model	MAE	RMSE	R^2
Naive baseline	5.946	9.053	-0.976
Linear Regression	4.337	6.466	-0.008
Random Forest	4.521	6.559	-0.038
XGBoost	4.629	6.827	-0.124

```
if model_name == "Naive Baseline":
    y_pred = X_test["..._lag_1"].values
elif model_name == "Linear Regression":
    model = LinearRegression()
    model.fit(X_train, y_train)
```

Advisor Logic Evidence

Decision flow

1. User selects location, sky view, time, weather, temperature, use case.
2. Advisor builds the same features used by the latency model.
3. Saved Linear Regression predicts next 15-minute latency only.
4. Jitter/download/upload use empirical or live estimates.
5. Traffic-light rules check all metrics against use-case thresholds.
6. Output: Suitable / Limited / Not Recommended + confidence.

Approximate forecast uncertainty

```
uncertainty = ±1.96 × residual_std
```

Shows a practical error range around the latency forecast.
Confidence improves when live measurements replace condition-based estimates.

Code from advisor.py

```
latency = saved_model.predict(X_lat)[0]
lat_uncert = 1.96 * SAVED_MODEL_RESIDUAL_STD

ratings = {
    uc: traffic_light(uc, latency, jitter, download, upload)
}
```

Confidence logic

```
if live_count >= 4 and latency_uncertainty <= 8:
    return "High"
if live_count >= 2 and latency_uncertainty <= 10:
    return "Medium"
return "Low"
```

Dashboard Logic Evidence

Dashboard modules

- Validated experiment windows are loaded and fixed.
- KPI cards calculate latest latency, jitter, download, upload and health score.
- Historical tabs show time-series, hourly patterns, weather and ISP comparisons.
- Forecasting tab compares models and shows next-interval prediction.
- Alerts convert forecasted values into user-readable warnings.

Health score code

```
def compute_health(latency, jitter, download, upload):
    score = 100
    if latency > 80: score -= 30
    elif latency > 50: score -= 15
    if jitter > 15: score -= 25
    if download < 20: score -= 20
    if upload < 10: score -= 15
    return score, label
```

Comparison logic

```
for metric in ["ping_avg_rtt_ms", "download_mbps", "upload_mbps", "ping_jitter_ms"]:
    compare Starlink mean vs Omantel/Awasr mean
```

Throughput investigation

What measured pattern looked like

- Fibre subscriptions were 500 Mbps, but automated test results stayed far lower.
- Omantel download: mean 36.2 Mbps, P95 94.7 Mbps.
- Awasr download: mean 27.2 Mbps.
- Awasr upload: mean 4.0 Mbps, P5 1.0 Mbps.
- Starlink upload: about 35 Mbps mean and 15-16 Mbps at P5.

From Speedtest CLI

- CLI/help shows server-related options such as:
  --servers (list nearest servers)
  --server-id=id (force a specific server)
  --selection-details (show server selection details)

This means the measured throughput depends on the selected test server and not only the access line.

From the logger code

```
res = run_cmd(["speedtest", "-f", "json", "--accept-license",
               "--accept-gdpr", "--progress=no"])
server = data.get("server", {})
server_name = server.get("name", None)
server_loc  = server.get("location", None)
```

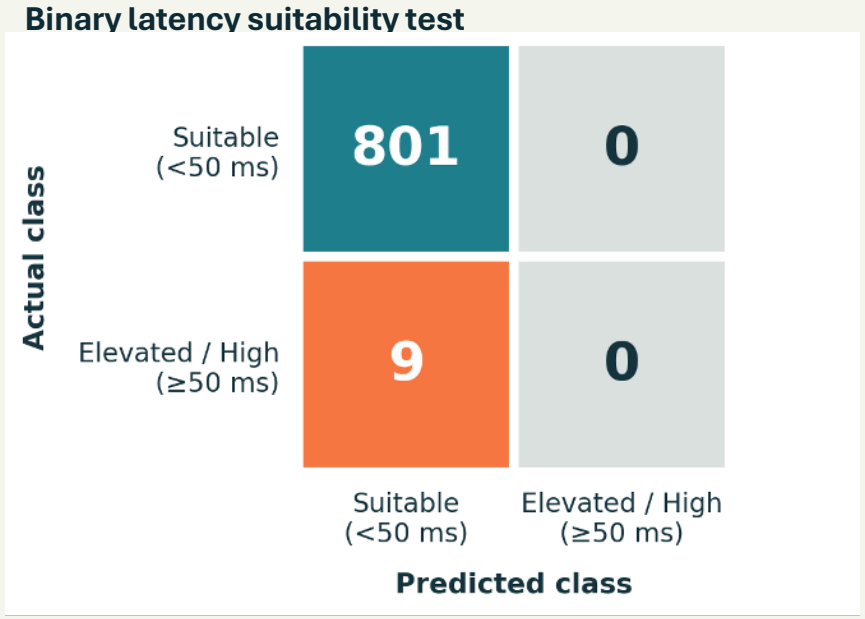
From the literature

- Speed-test studies say measured speed can be affected by test design, client/server implementation, and measurement infrastructure.
- MacMillan et al. (2022) specifically report cases of systematically under-performing Ookla servers.
- This supports treating low fibre throughput as a possible server/test-path limitation, not proof that the physical fibre line was only that speed.

Notes based on: logger code, thesis throughput results, official Starlink speed information, and MacMillan et al. (2022) on speed-test server behaviour.

Advisor Classification Check

The advisor combines the latency forecast with jitter, download, and upload thresholds for each use case.



98.9% overall accuracy

Accuracy is inflated by class imbalance: all 9 elevated/high cases were predicted as Suitable.

Measure	Result
Actual Suitable	801
Actual Elevated/High	9
Elevated/High recall	0.0%
False-negative rate	100.0%
Balanced accuracy	50.0%
Macro-F1	49.7%

Conclusion: useful for normal-condition guidance, but not a reliable latency-spike warning system yet.

Source: Thesis Ch. 4, Section 4.9.1 and Figure 22.

How Starlink Works

User terminal electronically tracks passing satellites

LEO altitude reduces latency compared with GEO

Laser links may relay traffic before reaching a gateway

Gateway connects the Starlink network to the terrestrial internet

1 **User Devices**

2 **Home Wi-Fi Router**

3 **Starlink Dish / User Terminal**

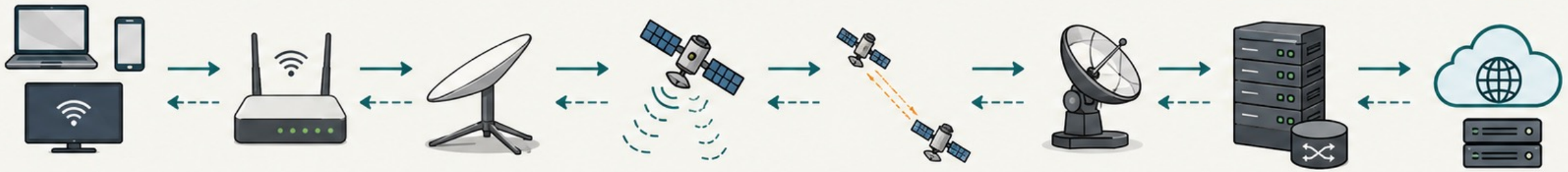
4 **Radio Uplink/Downlink to LEO Satellite**

5 **Inter-Satellite Laser Links (Optional)**

6 **Ground Gateway Station**

7 **Starlink Core / Terrestrial Backbone / Point of Presence**

8 **Public Internet / Cloud / Application Servers**



Laptop, phone, smart TV, etc.

1. Device traffic reaches the router
2. Router sends traffic to the dish
3. Dish exchanges data with LEO satellites
4. Traffic exits through a gateway to the internet

Source: SpaceX Starlink technology overview; adapted for thesis illustration.

Statistical Methods: Kruskal–Wallis & Spearman

Kruskal–Wallis test

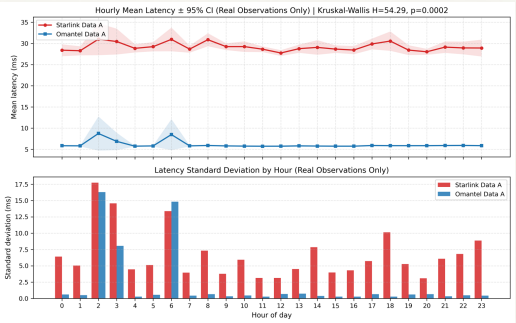
Non-parametric test for differences across k independent groups using ranks.

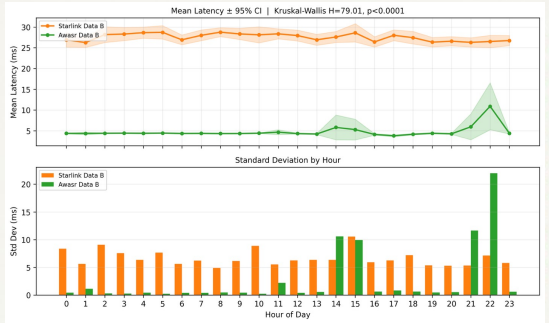
$$H = \left(\frac{12}{N(N+1)}\right) \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(N+1)$$

$$H_c = \frac{H}{\left[1 - \frac{\sum (t^3 - t)}{N^3 - N}\right]}$$

N = total observations
k = number of groups
n_j = size of group j
R_j = sum of ranks in group j
t = number of tied values

H_0: all group distributions are equal.
Approximate p–value from χ^2 with $df = k - 1$.





Spearman correlation

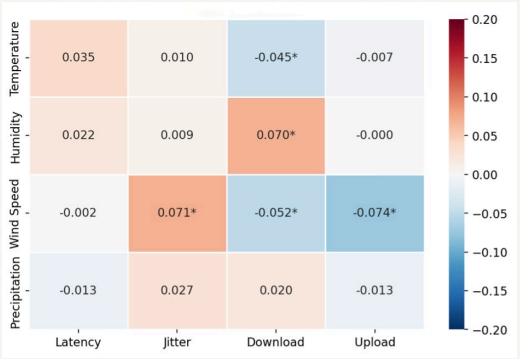
Non-parametric measure of monotonic association between two variables using ranks.

$$\rho_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

$$\rho_s = Corr(rank(X), rank(Y))$$

n = number of paired observations
d_i = difference between paired ranks

H_0: $\rho_s = 0$.
Used to test weather–network relationships.





Source: thesis statistical methodology; formulas adapted from standard non-parametric statistics references.